A Notation

For each shared upper-layer model, we equivalently regard the corresponding main and auxiliary clients as holding identical top model in the theoretical analysis. Let the global model at communication round t be denoted by $\mathbf{x}_t = (\mathbf{v}_t, \mathbf{u}_t)$, where $\mathbf{v}_t$ and $\mathbf{u}_t$ represent the bottom and top model, respectively. For device i, let $F_i(\mathbf{x})$ denote its local objective function with $\mathbf{x} = (\mathbf{v}, \mathbf{u})$, and define the global objective function as

$$F(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} \Big(F_i(\mathbf{x}_i) \Big). \tag{21}$$

At global round t, device i performs K local iterations indexed by $k = 1, \ldots, K$.

We denote the local model at the k-th local iteration by $\mathbf{x}_{i,t}^k = (\mathbf{v}_{i,t}^k, \mathbf{u}_{i,t}^k)$, and let $\boldsymbol{\xi}_{i,t}^k$ denote the random sample drawn at this local state. The stochastic gradients of device i with respect to $\mathbf{v}$ and $\mathbf{u}$ are denoted by $\mathbf{h}_i(\mathbf{x}_{i,t}^k; \boldsymbol{\xi}_{i,t}^k)$ and $\mathbf{g}_i(\mathbf{x}_{i,t}^k; \boldsymbol{\xi}_{i,t}^k)$, respectively.

The corresponding true gradients are denoted by $\nabla_{\mathbf{v}} F_i(\mathbf{x}_{i,t}^k)$ and $\nabla_{\mathbf{u}} F_i(\mathbf{x}_{i,t}^k)$ for the local objective, and by $\nabla_{\mathbf{v}} F(\mathbf{x}_t)$ and $\nabla_{\mathbf{u}} F(\mathbf{x}_t)$ for the global objective.

B Convergence Analysis

B.1 Pair-level cumulative update.

The N devices are partitioned into P collaborative pairs, where $N = 2P$. For the p-th pair, denote the main-side and auxiliary-side bottom model by $\mathbf{v}_{p,t}^m$ and $\mathbf{v}_{p,t}^a$, respectively, and the shared upper-layer parameter by $\mathbf{u}_{p,t}$. We define the pair-level cumulative update at round t as

$$Q_{p,t} := \left(Q_{p,t}^{m,v}, Q_{p,t}^{a,v}, Q_{p,t}^{u} \right), \tag{22}$$

where

$$Q_{p,t}^{m,v} := \sum_{k=1}^{K} h_{p,t}^{m,k}, \qquad Q_{p,t}^{a,v} := \sum_{k=1}^{K} h_{p,t}^{a,k}, \tag{23}$$

and

$$Q_{p,t}^{u} := \sum_{k=1}^{K} \big(g_{p,t}^{a,k} + g_{p,t}^{m,k} - r_{p,t}^k \big). \tag{24}$$

where

$$r_{p,t}^k := g_{p,t}^{m,k} - g_{p,t}^{m,k+\frac{1}{2}} \tag{25}$$

Accordingly, the global update can be written as

$$\mathbf{x}_{t+1} - \mathbf{x}_t = -\frac{\eta}{N} \sum_{p=1}^{P} Q_{p,t}. \tag{26}$$

For convenience, define the conditional mean

$$\bar{Q}_{p,t} := \mathbb{E}[Q_{p,t} \mid \mathbf{x}_t]. \tag{27}$$

B.2 Non-Convex Convergence Analysis

Lemma 1 (Pair-level mean decomposition). *Under Assumption 3, for each collaborative pair p and round t, the conditional mean $\bar{Q}_{p,t}$ satisfies*

$$\bar{Q}_{p,t} = \Big(K\nabla_{\mathbf{v}} F(\mathbf{x}_t),\ K\nabla_{\mathbf{v}} F(\mathbf{x}_t),\ 2K\nabla_{\mathbf{u}} F(\mathbf{x}_t)\Big) - \bar{R}_{p,t}, \tag{28}$$

where

$$\bar{R}_{p,t} := \left(0,\ 0,\ \sum_{k=1}^{K} \bar{r}_{p,t}^{k}\right), \qquad \bar{r}_{p,t}^{k} := \mathbb{E}[r_{p,t}^{k} \mid \mathbf{x}_t]. \tag{29}$$

Lemma 2 (Pair-level stochastic deviation bound). *For each collaborative pair p and round t, define*

$$\epsilon_{p,t} := Q_{p,t} - \bar{Q}_{p,t}. \tag{30}$$

Under assumptions 1 2 3, we have

$$\mathbb{E}\|\epsilon_{p,t}\|^2 \le 2K\sigma_v^2 + 6K\sigma_u^2 + 6KL^2\eta^2(\sigma_u^2 + G_u^2). \tag{31}$$

Lemma 3 (Pair-level residual mean bound). *For each collaborative pair p and round t, let*

$$\bar{R}_{p,t} := \left(0,\ 0,\ \sum_{k=1}^{K} \bar{r}_{p,t}^{k}\right), \qquad \bar{r}_{p,t}^{k} := \mathbb{E}[r_{p,t}^{k} \mid \mathbf{x}_t]. \tag{32}$$

Under assumptions 1 2 3, we have

$$\mathbb{E}\|\bar{R}_{p,t}\|^2 \le 2K^2L^2\eta^2(\sigma_u^2 + G_u^2). \tag{33}$$

Moreover, for any $\alpha_t > 0$,

$$\mathbb{E}\langle \nabla_u F(\mathbf{x}_t), \bar{R}_{p,t}^{u} \rangle \le \frac{\alpha_t}{2}\mathbb{E}\|\nabla_u F(\mathbf{x}_t)\|^2 + \frac{K^2L^2\eta^2}{\alpha_t}(\sigma_u^2 + G_u^2). \tag{34}$$

Proof of Theorem 1

Proof. By assumption 1 and the L-smoothness of $F(\cdot)$,

$$F(\mathbf{x}_{t+1}) \le F(\mathbf{x}_t) + \langle \nabla F(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t \rangle + \frac{L}{2}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2. \tag{35}$$

Using the pair-level update rule

$$\mathbf{x}_{t+1} - \mathbf{x}_t = -\frac{\eta}{N}\sum_{p=1}^{P} Q_{p,t}, \tag{36}$$

and taking expectation, we obtain

$$\mathbb{E}[F(\mathbf{x}_{t+1})] \leq \mathbb{E}[F(\mathbf{x}_t)] - \frac{\eta}{N}\sum_{p=1}^{P}\mathbb{E}\langle \nabla F(\mathbf{x}_t), Q_{p,t}\rangle + \frac{L}{2}\mathbb{E}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2. \tag{37}$$

Write $Q_{p,t} = \bar{Q}_{p,t} + (Q_{p,t} - \bar{Q}_{p,t})$, where $\bar{Q}_{p,t} = \mathbb{E}[Q_{p,t} \mid \mathbf{x}_t]$. Since $\mathbb{E}[Q_{p,t} - \bar{Q}_{p,t} \mid \mathbf{x}_t] = 0$, we have

$$\mathbb{E}\langle \nabla F(\mathbf{x}_t), Q_{p,t}\rangle = \mathbb{E}\langle \nabla F(\mathbf{x}_t), \bar{Q}_{p,t}\rangle. \tag{38}$$

By Lemma 1,

$$\bar{Q}_{p,t} = \Big(K\nabla_v F(\mathbf{x}_t),\ K\nabla_v F(\mathbf{x}_t),\ 2K\nabla_u F(\mathbf{x}_t)\Big) - \bar{R}_{p,t}. \tag{39}$$

Hence,

$$\langle \nabla F(\mathbf{x}_t), \bar{Q}_{p,t}\rangle = 2K\|\nabla F(\mathbf{x}_t)\|^2 - \langle \nabla_u F(\mathbf{x}_t), \bar{R}^u_{p,t}\rangle. \tag{40}$$

Applying Lemma 3 with $\alpha_t = K$ yields

$$\mathbb{E}\langle \nabla F(\mathbf{x}_t), \bar{Q}_{p,t}\rangle \geq \frac{3K}{2}\mathbb{E}\|\nabla F(\mathbf{x}_t)\|^2 - KL^2\eta^2(\sigma_u^2 + G_u^2). \tag{41}$$

Summing over $p = 1, \ldots, P$ and using $N = 2P$, we obtain

$$-\frac{\eta}{N}\sum_{p=1}^{P}\mathbb{E}\langle \nabla F(\mathbf{x}_t), \bar{Q}_{p,t}\rangle \leq -\frac{3K}{4}\eta\,\mathbb{E}\|\nabla F(\mathbf{x}_t)\|^2 + \frac{KL^2}{2}\eta^3(\sigma_u^2 + G_u^2). \tag{42}$$

Next, for the second-order term,

$$\mathbb{E}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 = \frac{\eta^2}{N^2}\mathbb{E}\left\|\sum_{p=1}^{P} Q_{p,t}\right\|^2. \tag{43}$$

Writing again $Q_{p,t} = \bar{Q}_{p,t} + \epsilon_{p,t}$ with $\epsilon_{p,t} := Q_{p,t} - \bar{Q}_{p,t}$, and noting that $\mathbb{E}[\epsilon_{p,t} \mid \mathbf{x}_t] = 0$, the cross term between the conditional mean part and the stochastic deviation part vanishes. Hence,

$$\mathbb{E}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 = \frac{\eta^2}{N^2}\mathbb{E}\left\|\sum_{p=1}^{P} \bar{Q}_{p,t}\right\|^2 + \frac{\eta^2}{N^2}\mathbb{E}\left\|\sum_{p=1}^{P} \epsilon_{p,t}\right\|^2. \tag{44}$$

For the conditional mean term, by Lemma 1, we have

$$\sum_{p=1}^{P}\bar{Q}_{p,t} = P\big(K\nabla_v F(\mathbf{x}_t),\, K\nabla_v F(\mathbf{x}_t),\, 2K\nabla_u F(\mathbf{x}_t)\big) - \sum_{p=1}^{P}\bar{R}_{p,t}. \tag{45}$$

Hence, by $\|a-b\|^2 \le 2\|a\|^2 + 2\|b\|^2$,

$$\begin{aligned}\mathbb{E}\left\|\sum_{p=1}^{P} \bar{Q}_{p,t}\right\|^2 \le 2P^2 \left\|\left(K\nabla_v F(\mathbf{x}_t),\, K\nabla_v F(\mathbf{x}_t),\, 2K\nabla_u F(\mathbf{x}_t)\right)\right\|^2 \\ + 2\mathbb{E}\left\|\sum_{p=1}^{P} \bar{R}_{p,t}\right\|^2.\end{aligned} \tag{46}$$

Moreover,

$$\begin{aligned}&\left\|\left(K\nabla_v F(\mathbf{x}_t),\, K\nabla_v F(\mathbf{x}_t),\, 2K\nabla_u F(\mathbf{x}_t)\right)\right\|^2 \\ =&2K^2\|\nabla_v F(\mathbf{x}_t)\|^2 + 4K^2\|\nabla_u F(\mathbf{x}_t)\|^2 \\ \le&4K^2\|\nabla F(\mathbf{x}_t)\|^2.\end{aligned} \tag{47}$$

Therefore,

$$\mathbb{E}\left\|\sum_{p=1}^{P} \bar{Q}_{p,t}\right\|^2 \le 8P^2K^2\,\mathbb{E}\|\nabla F(\mathbf{x}_t)\|^2 + 2\mathbb{E}\left\|\sum_{p=1}^{P} \bar{R}_{p,t}\right\|^2. \tag{48}$$

Using

$$\left\|\sum_{p=1}^{P} \bar{R}_{p,t}\right\|^2 \le P\sum_{p=1}^{P} \|\bar{R}_{p,t}\|^2 \tag{49}$$

and Lemma 3, we further obtain

$$\mathbb{E}\left\|\sum_{p=1}^{P} \bar{Q}_{p,t}\right\|^2 \le 8P^2K^2\,\mathbb{E}\|\nabla F(\mathbf{x}_t)\|^2 + 4P^2K^2L^2\eta^2(\sigma_u^2 + G_u^2). \tag{50}$$

Under pair-wise independent sampling, the cross terms between different collaborative pairs vanish. Applying Lemma 2 and Lemma 3, we obtain

$$\mathbb{E}\|\mathbf{x}_{t+1}-\mathbf{x}_t\|^2 \le 2K^2\eta^2\mathbb{E}\|\nabla F(\mathbf{x}_t)\|^2 + \frac{\eta^2}{N}(K\sigma_v^2 + 3K\sigma_u^2) + \eta^4L^2(\sigma_u^2+G_u^2)\left(K^2+\frac{3K}{N}\right). \tag{51}$$

Substituting (42) and (51) into the one-step descent inequality gives

$$\begin{aligned}\mathbb{E}[F(\mathbf{x}_{t+1})] \le \mathbb{E}[F(\mathbf{x}_t)] - \left(\frac{3K}{4}\eta - LK^2\eta^2\right)\mathbb{E}\|\nabla F(\mathbf{x}_t)\|^2 \\ + \frac{LK}{2N}\eta^2(\sigma_v^2 + 3\sigma_u^2) + \frac{KL^2}{2}\eta^3(\sigma_u^2 + G_u^2) \\ + \frac{L^3}{2}\eta^4(\sigma_u^2 + G_u^2)\left(K^2 + \frac{3K}{N}\right).\end{aligned} \tag{52}$$

When $LK^2\eta^2 \le \frac{1}{4}K\eta$, that is $\eta \le \frac{1}{4LK}$, we have

$$\mathbb{E}[F(\mathbf{x}_{t+1})] \le \mathbb{E}[F(\mathbf{x}_t)] - \frac{K}{2}\eta\,\mathbb{E}\|\nabla F(\mathbf{x}_t)\|^2 + \eta K C_t, \tag{53}$$

where

$$C_t = \frac{L}{2N}\eta(\sigma_v^2 + 3\sigma_u^2) + \frac{L^2}{2}\eta^2(\sigma_u^2 + G_u^2) + \frac{L^3}{2}\eta^3(\sigma_u^2 + G_u^2)\left(K + \frac{3}{N}\right). \quad (54)$$

Summing the above inequality from $t = 0$ to $T - 1$ and using the lower boundedness of $F(\cdot)$ yield

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla F(\mathbf{x}_t)\|^2 \leq \frac{2(F(\mathbf{x}_0) - F(\mathbf{x}_T))}{\eta K T} + 2C_t. \quad (55)$$

Let $\eta = \sqrt{\frac{2(F(\mathbf{x_0})-F(\mathbf{x}_T))N}{(\sigma_v^2+3\sigma_u^2)LKT}}$ into Eq. (54)

we analyze the asymptotic order of each error component in C_t.

The first term, which encapsulates gradient noise and local bias, scales as $\Theta(\eta) = \Theta(T^{-1/2})$, which matches the order of the leading term in Eq. (55).

The second term, induced by second-order effects from local iterations, scales as $\Theta(\eta^2) = \Theta(T^{-1})$, a higher-order term that decays faster than the leading convergence rate.

The third term, which is explicitly introduced by the collaborative alternating update mechanism of SFLMAC, scales as $\Theta(\eta^3) = \Theta(T^{-3/2})$, a negligible higher-order term that vanishes much more rapidly than $O(T^{-1/2})$ as the number of communication rounds T increases.

For sufficiently large T, the overall upper bound in Eq. (55) is asymptotically dominated by the leading-order term $O(T^{-1/2})$, while all higher-order terms—including those introduced by the collaborative update mechanism—become negligible. This leads to the result:

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla F(\mathbf{x}_t)\|^2 \preceq O\left(\frac{1}{\sqrt{T}}\right),$$

where $\preceq$ denotes the standard order inequality.

Therefore, SFLMAC preserves the standard $O(1/\sqrt{T})$ convergence rate of non-convex stochastic first-order methods.

The collaborative alternating update only contributes higher-order terms that decay faster than the leading rate and thus does not affect the asymptotic convergence behavior of the algorithm, validating the theoretical convergence guarantee of the proposed method.

This completes the proof.

B.3 Proof of Lemma

Proof of Lemma 1

Proof. By the definition of $Q_{p,t}$ and linearity of conditional expectation,

$$\bar{Q}_{p,t} = \left(\sum_{k=1}^{K}\mathbb{E}[h_{p,t}^{m,k} \mid \mathbf{x}_t],\ \sum_{k=1}^{K}\mathbb{E}[h_{p,t}^{a,k} \mid \mathbf{x}_t],\ \sum_{k=1}^{K}\mathbb{E}[g_{p,t}^{m,k} + g_{p,t}^{a,k} - r_{p,t}^{k} \mid \mathbf{x}_t]\right). \quad (56)$$

Under Assumption 3, the stochastic gradients are unbiased estimators of the corresponding global gradients at $\mathbf{x}_t$, i.e.,

$$\mathbb{E}[h_{p,t}^{m,k} \mid \mathbf{x}_t] = \nabla_v F(\mathbf{x}_t), \quad \mathbb{E}[h_{p,t}^{a,k} \mid \mathbf{x}_t] = \nabla_v F(\mathbf{x}_t), \tag{57}$$

and

$$\mathbb{E}[g_{p,t}^{m,k} \mid \mathbf{x}_t] = \nabla_u F(\mathbf{x}_t), \quad \mathbb{E}[g_{p,t}^{a,k} \mid \mathbf{x}_t] = \nabla_u F(\mathbf{x}_t). \tag{58}$$

Substituting the above identities into the definition of $\bar{Q}_{p,t}$ yields (28).

This completes the proof.

Proof of Lemma2

Proof. By the definitions of $Q_{p,t}$ and $\bar{Q}_{p,t}$,

$$\epsilon_{p,t} = \left(\sum_{k=1}^{K} (h_{p,t}^{m,k} - \nabla_v F(\mathbf{x}_t)),\ \sum_{k=1}^{K} (h_{p,t}^{a,k} - \nabla_v F(\mathbf{x}_t)),\ \sum_{k=1}^{K} \zeta_{p,t}^{k} \right), \tag{59}$$

where

$$\zeta_{p,t}^{k} := (g_{p,t}^{m,k} - \nabla_u F(\mathbf{x}_t)) + (g_{p,t}^{a,k} - \nabla_u F(\mathbf{x}_t)) - (r_{p,t}^{k} - \bar{r}_{p,t}^{k}). \tag{60}$$

Hence,

$$\begin{aligned} \mathbb{E}\|\epsilon_{p,t}\|^2 = \mathbb{E}\left\| \sum_{k=1}^{K} (h_{p,t}^{m,k} - \nabla_v F(\mathbf{x}_t)) \right\|^2 + \mathbb{E}\left\| \sum_{k=1}^{K} (h_{p,t}^{a,k} - \nabla_v F(\mathbf{x}_t)) \right\|^2 \\ + \mathbb{E}\left\| \sum_{k=1}^{K} \zeta_{p,t}^{k} \right\|^2. \end{aligned} \tag{61}$$

Since the samples are independently drawn from the same distribution $\mathcal{D}$, the cross terms between different local steps vanish. Therefore,

$$\mathbb{E}\left\| \sum_{k=1}^{K} (h_{p,t}^{m,k} - \nabla_v F(\mathbf{x}_t)) \right\|^2 = \sum_{k=1}^{K} \mathbb{E}\|h_{p,t}^{m,k} - \nabla_v F(\mathbf{x}_t)\|^2 \le K\sigma_v^2, \tag{62}$$

and similarly,

$$\mathbb{E}\left\| \sum_{k=1}^{K} (h_{p,t}^{a,k} - \nabla_v F(\mathbf{x}_t)) \right\|^2 \le K\sigma_v^2. \tag{63}$$

Since each $\zeta_{p,t}^{k}$ is formed from the k-th independent sample only and has zero conditional mean given x_t, the cross terms between different local steps vanish. For the upper-layer block, we also have

$$\mathbb{E}\left\| \sum_{k=1}^{K} \zeta_{p,t}^{k} \right\|^2 = \sum_{k=1}^{K} \mathbb{E}\|\zeta_{p,t}^{k}\|^2. \tag{64}$$

Using $\|a+b-c\|^2 \le 3\|a\|^2 + 3\|b\|^2 + 3\|c\|^2$, we obtain

$$\mathbb{E}\|\zeta_{p,t}^k\|^2 \le 3\mathbb{E}\|g_{p,t}^{m,k} - \nabla_u F(\mathbf{x}_t)\|^2 + 3\mathbb{E}\|g_{p,t}^{a,k} - \nabla_u F(\mathbf{x}_t)\|^2 + 3\mathbb{E}\|r_{p,t}^k - \bar{r}_{p,t}^k\|^2. \quad (65)$$

By assumption 3,

$$\mathbb{E}\|g_{p,t}^{m,k} - \nabla_u F(\mathbf{x}_t)\|^2 \le \sigma_u^2, \qquad \mathbb{E}\|g_{p,t}^{a,k} - \nabla_u F(\mathbf{x}_t)\|^2 \le \sigma_u^2. \quad (66)$$

Moreover, by variance decomposition,

$$\mathbb{E}\|r_{p,t}^k - \bar{r}_{p,t}^k\|^2 \le \mathbb{E}\|r_{p,t}^k\|^2. \quad (67)$$

By Lemma 3, we have:

$$\mathbb{E}\|r_{p,t}^k\|^2 \le 2L^2\eta^2(\sigma_u^2 + G_u^2), \quad (68)$$

and consequently,

$$\mathbb{E}\|\zeta_{p,t}^k\|^2 \le 6\sigma_u^2 + 6L^2\eta^2(\sigma_u^2 + G_u^2). \quad (69)$$

Summing over $k = 1, \ldots, K$ yields

$$\mathbb{E}\left\|\sum_{k=1}^{K} \zeta_{p,t}^k\right\|^2 \le K\Big(6\sigma_u^2 + 6L^2\eta^2(\sigma_u^2 + G_u^2)\Big). \quad (70)$$

Combining the above estimates proves (31).

This completes the proof.

Proof of Lemma 3

Proof. By definition,

$$\|\bar{R}_{p,t}\|^2 = \left\|\sum_{k=1}^{K} \bar{r}_{p,t}^k\right\|^2 \le K\sum_{k=1}^{K}\|\bar{r}_{p,t}^k\|^2. \quad (71)$$

Taking expectation on both sides yields

$$\mathbb{E}\|\bar{R}_{p,t}\|^2 \le K\sum_{k=1}^{K}\mathbb{E}\|\bar{r}_{p,t}^k\|^2. \quad (72)$$

Since $\bar{r}_{p,t}^k = \mathbb{E}[r_{p,t}^k \mid \mathbf{x}_t]$, Jensen's inequality gives

$$\|\bar{r}_{p,t}^k\|^2 \le \mathbb{E}\big[\|r_{p,t}^k\|^2 \mid \mathbf{x}_t\big]. \quad (73)$$

Hence,

$$\mathbb{E}\|\bar{r}_{p,t}^k\|^2 \le \mathbb{E}\|r_{p,t}^k\|^2. \quad (74)$$

Therefore,

$$\mathbb{E}\|\bar{R}_{p,t}\|^2 \le K\sum_{k=1}^{K}\mathbb{E}\|r_{p,t}^k\|^2. \quad (75)$$

Since the residual term is induced by the upper half-step update, by the Lipschitz continuity of the upper-layer stochastic gradient we have

$$\|r^k_{p,t}\|^2 \le L^2\eta^2\|g^k_{p,\mathrm{partner}}\|^2. \tag{76}$$

Thus,

$$\mathbb{E}\|r^k_{p,t}\|^2 \le L^2\eta^2\mathbb{E}\|g^k_{p,\mathrm{partner}}\|^2. \tag{77}$$

Using

$$g^k_{p,\mathrm{partner}} = (g^k_{p,\mathrm{partner}} - \nabla_u F(\mathbf{x}_t)) + \nabla_u F(\mathbf{x}_t) \tag{78}$$

and $\|a+b\|^2 \le 2\|a\|^2 + 2\|b\|^2$, we obtain

$$\mathbb{E}\|g^k_{p,\mathrm{partner}}\|^2 \le 2\mathbb{E}\|g^k_{p,\mathrm{partner}} - \nabla_u F(\mathbf{x}_t)\|^2 + 2\mathbb{E}\|\nabla_u F(\mathbf{x}_t)\|^2. \tag{79}$$

By assumptions 2 3,

$$\mathbb{E}\|g^k_{p,\mathrm{partner}}\|^2 \le 2\sigma_u^2 + 2G_u^2. \tag{80}$$

Hence,

$$\mathbb{E}\|r^k_{p,t}\|^2 \le 2L^2\eta^2(\sigma_u^2 + G_u^2). \tag{81}$$

Substituting into the previous bound yields

$$\mathbb{E}\|\bar{R}_{p,t}\|^2 \le K\sum_{k=1}^{K} 2L^2\eta^2(\sigma_u^2 + G_u^2) = 2K^2L^2\eta^2(\sigma_u^2 + G_u^2). \tag{82}$$

This proves (33).

Finally, by Young's inequality,

$$\langle \nabla_u F(\mathbf{x}_t), \bar{R}^u_{p,t}\rangle \le \frac{\alpha_t}{2}\|\nabla_u F(\mathbf{x}_t)\|^2 + \frac{1}{2\alpha_t}\|\bar{R}^u_{p,t}\|^2. \tag{83}$$

Taking expectation and using (33) gives

$$\mathbb{E}\langle \nabla_u F(\mathbf{x}_t), \bar{R}^u_{p,t}\rangle \le \frac{\alpha_t}{2}\mathbb{E}\|\nabla_u F(\mathbf{x}_t)\|^2 + \frac{K^2L^2\eta^2}{\alpha_t}(\sigma_u^2 + G_u^2). \tag{84}$$

This completes the proof.

C Optimal client pairing selection

Theorem 2. *Let $N = 2P$ clients have computing capacities sorted in ascending order as*

$$x_1 \le x_2 \le \cdots \le x_N.$$

Assume that each client initially owns a task of size M, and for each pair (i,j) with $x_i \le x_j$, the auxiliary client i is allowed to offload an arbitrary portion of its task to the main client j. Then, for a fixed pair (i,j), the minimum completion time is

$$T^\star_{ij} = \frac{2M}{x_i + x_j},$$

achieved by the balanced split $m_i^\star = \frac{2Mx_i}{x_i+x_j}$

$$\frac{m_i^*}{x_i} = \frac{2M - m_i^*}{x_j}$$

Consequently, the global pairing problem

$$\min_{\Pi} \max_{(i,j)\in\Pi} T_{ij}^\star$$

is equivalent to

$$\max_{\Pi} \min_{(i,j)\in\Pi} (x_i + x_j).$$

Among all pairing schemes, the strongest-weakest pairing

$$(1, N), (2, N-1), \ldots, (P, P+1)$$

is optimal.

Proof. For any pair (i, j) with $x_i \le x_j$, let $m_i \in [0, M]$ denote the workload processed by client i. Then client j processes workload $2M - m_i$, and the pair completion time is

$$T_{ij}(m_i) = \max\left\{\frac{m_i}{x_i}, \frac{2M - m_i}{x_j}\right\}.$$

Since $\frac{m_i}{x_i}$ is increasing in m_i and $\frac{2M-m_i}{x_j}$ is decreasing in m_i, the minimum is attained when the two terms are equal:

$$\frac{m_i}{x_i} = \frac{2M - m_i}{x_j}.$$

Solving gives

$$m_i^\star = \frac{2Mx_i}{x_i + x_j}$$

and hence

$$T_{ij}^\star = \frac{2M}{x_i + x_j}.$$

Therefore, for any pairing policy Π,

$$T(\Pi) = \max_{(i,j)\in\Pi} T_{ij}^\star = \max_{(i,j)\in\Pi} \frac{2M}{x_i + x_j} = \frac{2M}{\min_{(i,j)\in\Pi}(x_i + x_j)}.$$

Thus, minimizing $T(\Pi)$ is equivalent to maximizing

$$\min_{(i,j)\in\Pi} (x_i + x_j).$$

We now prove that the strongest-weakest pairing

$$\Pi^\star = \{(1, N), (2, N-1), \ldots, (P, P+1)\}$$

maximizes the minimum pair sum.

Let Π be any pairing. Since client 1 must be paired with some client k, we have

$$x_1 + x_k \leq x_1 + x_N.$$

Hence,

$$\min_{(i,j)\in\Pi}(x_i + x_j) \leq x_1 + x_k \leq x_1 + x_N.$$

So no pairing can have minimum pair sum larger than $x_1 + x_N$.

Now we show that there exists an optimal pairing containing the pair $(1, N)$. Suppose in some optimal pairing, client 1 is paired with a and client N is paired with b, where $a \neq N$ and $b \neq 1$. Then these two pairs are $(1, a)$ and (b, N), with $x_1 \leq x_b \leq x_a \leq x_N$ after relabeling if necessary. Replace them by $(1, N)$ and (a, b). The new pair sums are

$$x_1 + x_N, \qquad x_a + x_b.$$

Compared with the original pair sums

$$x_1 + x_a, \qquad x_b + x_N,$$

we have

$$x_1 + x_N \geq x_1 + x_a, \qquad x_a + x_b \geq x_1 + x_a$$

Therefore, the minimum of the two new pair sums is no smaller than the minimum of the two original pair sums. Hence this replacement does not decrease the minimum pair sum. So there always exists an optimal pairing containing $(1, N)$.

After fixing the pair $(1, N)$, the remaining clients are

$$x_2 \leq x_3 \leq \cdots \leq x_{N-1},$$

which form the same problem of size $N - 2$. Repeating the same argument recursively shows that there exists an optimal pairing containing

$$(1, N), (2, N-1), \ldots, (P, P+1).$$

Therefore, the strongest-weakest pairing maximizes the minimum pair sum, and equivalently minimizes the per-round completion time.

This completes the proof.